

DARSHAN R

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SUMMARY

Computer Science undergraduate at IIIT Sri City focused on AI product development, retrieval-augmented generation, and applied machine learning systems. Built end-to-end AI workflows spanning data ingestion, prompting, retrieval, reranking, and citation-grounded generation. Interested in narrative AI systems, prompt design, and shipping AI features that work reliably at product scale.

EDUCATION

Indian Institute of Information Technology (IIIT), Sricity <i>Bachelor of Technology – Computer Science and Engineering; CGPA: 9.13</i>	Andhra Pradesh, India 2023 – 2027
Narayana E-Techno School <i>Higher Secondary Education; Grade: 94% Percentile: 96%</i>	India 2017 – 2023

EXPERIENCE

Research Intern <i>National Institute of Technology (NIT), Trichy</i>	May 2025 – June 2025 Tamil Nadu, India
- Developed an autonomous lane-changing agent using Deep Q-Networks (DQN) in Highway-env to study sequential decision-making under dynamic traffic conditions - Designed reward shaping and evaluated policies over 5000+ episodes using Double DQN with Prioritized Experience Replay - Collaborated with research teammates to document methods and empirical findings for a paper under review	
Core Member – Nexsync Smart Transportation Club <i>IIIT Sricity</i>	Sep 2024 – Present Andhra Pradesh, India
- Organized student hackathons and coordinated execution with technical and non-technical teams - Built a trip advisor web product with Node.js and React for personalized planning workflows	

AI PRODUCT & TECHNICAL PROJECTS

Ask My Docs – Citation-Aware RAG System <i>Python, FastAPI, Streamlit, ChromaDB</i>	2025
- Built a production-style RAG pipeline combining BM25, vector retrieval, hybrid fusion, and cross-encoder reranking for accurate document Q&A - Implemented citation-aware prompting with validation and retry logic to enforce grounded answers and reduce hallucinations - Added multi-LLM support (OpenAI/Ollama/Gemini), document upload lifecycle, and evaluation-ready architecture for iterative product improvement	
RAG Code Workspace – AI Assistant for Codebases <i>Next.js, React, Node.js, Express, Chroma</i>	2025
- Developed a VS Code-like AI workspace to upload/index repositories, ask codebase questions, and generate file-level explanations with citations - Engineered backend indexing and chunking pipelines with dependency-aware retrieval for better context quality - Designed product workflows for upload management, chat history persistence, and citation click-through into exact code ranges	
SafeLane – Reinforcement Learning for Autonomous Driving <i>Python, TensorFlow, Gymnasium</i>	2025
- Implemented a safety-aware DQN agent with adaptive reward formulation and scenario-based evaluation in Highway-env - Analyzed training behavior and reward trends across 5000 episodes to improve policy stability and reduce collisions	
Stampede Region Prediction – Crowd Risk Analytics <i>Python, PyTorch, OpenCV</i>	2025
- Built a deep learning pipeline for crowd density estimation and stampede risk detection from image and video streams - Implemented temporal risk analysis over grid regions to detect rapid accumulation and high-risk crowd movement patterns	

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript, TypeScript
AI/ML: PyTorch, TensorFlow, OpenCV, Reinforcement Learning, CNNs, RAG Pipelines
LLM & Retrieval: Prompt Engineering, Hybrid Retrieval, Reranking, Citation Grounding, ChromaDB
Product & Backend: FastAPI, Node.js, Express.js, REST APIs, Workflow Design
Frontend: React.js, Next.js, HTML, CSS, Tailwind CSS, Streamlit
Tools: Git, GitHub, Vite, Docker, Ollama

LEADERSHIP & ACTIVITIES

Languages: English, Telugu, Tamil (Native)
Awards: Football Winner at Abhisarga (IIIT Sricity) 2024 | Football Runner-Up at Sanyog (IIT Tirupati) 2024
Leadership: Head Boy, Narayana E-Techno School (Academic Year 2022–23)